SUPPLEMENTARY MATERIAL TO "MODELING GAP TIMES BETWEEN RECURRENT EVENTS: AN APPLICATION TO A MALARIA COHORT STUDY IN UGANDA"

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A. Data structure for fitting the models

Using linear mixed random effects model style of data input, each participant is represented by several rows of data depending on the number of visits they had. To highlight important features of the data structure considered in our application, we present observations of one selected participant (subject id = 2093) with 24 rows (corresponding to 24 visits during the study period). The observations of this subject are shown in Table A.1.

In these data, id and hhid are, respectively, participant and household identification numbers; malaria and parasitaemia equal to one correspond to the malaria condition, that is, symptomatic infection; malaria zero and parasitaemia equal to one correspond to asymptomatic infection; and Malaria and parasitaemia equal to zero indicate that the participant is free from infection. This participant, for example, had five malaria parasite infections, all of which were symptomatic.

Additional covariates include siteid (1 = Walukuba, 2 = Kihihi, 3 = Nagongera), previous use of Artemether-Lumefantrine (AL) treatment (usedAL = 1: participant received treatment with AL antimalaria medication, usedAL = 0: participant not received treatment) and age. Table A.2 displays the gap times between recurrent parasite infections. A pair of variables $[u_{ijk}, v_{ijk}]$ is used to define the risk time interval, where $u_{ijk} \in [u_{l_{ijk}},\ u_{r_{ijk}}]$ and $v_{ijk} \in [v_{l_{ijk}}\ v_{r_{ijk}}]$.

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Table A.1. PRISM study data: Observed data for subject id = 2093 with the following information: household identification number (hhid), study area (siteid), cohort age (cohortage), time visits (calendar time), malaria, parasitaemia, and AL treatment status.

hhid	id	siteid	age	cohortage	calendar-time	malaria	parasitaemia	usedAL
349004602	2093	2	5.920548	5.920548	0.06849315	0	0	0
349004602	2093	2	5.920548	6.090411	0.23835616	1	1	0
349004602	2093	2	5.920548	6.145205	0.29315068	0	0	1
349004602	2093	2	5.920548	6.375342	0.52185792	0	0	1
349004602	2093	2	5.920548	6.482192	0.6284153	0	0	1
349004602	2093	2	5.920548	6.517808	0.66393443	0	0	1
349004602	2093	2	5.920548	6.605479	0.75136612	0	0	1
349004602	2093	2	5.920548	6.849315	0.99453552	0	0	1
349004602	2093	2	5.920548	6.978082	1.12295082	1	1	1
349004602	2093	2	5.920548	7.09863	1.2431694	0	0	1
349004602	2093	2	5.920548	7.317808	1.46575342	0	0	1
349004602	2093	2	5.920548	7.347945	1.49589041	0	0	1
349004602	2093	2	5.920548	7.424657	1.57260274	0	0	1
349004602	2093	2	5.920548	7.580822	1.72876712	0	0	1
349004602	2093	2	5.920548	7.676713	1.82465753	1	1	1
349004602	2093	2	5.920548	7.849315	1.99726027	0	0	1
349004602	2093	2	5.920548	7.873972	2.02191781	0	0	1
349004602	2093	2	5.920548	7.887671	2.03561644	0	0	1
349004602	2093	2	5.920548	8.126027	2.2739726	0	0	1
349004602	2093	2	5.920548	8.369863	2.51092896	0	0	1
349004602	2093	2	5.920548	8.438356	2.57923497	1	1	1
349004602	2093	2	5.920548	8.616438	2.7568306	0	0	1
349004602	2093	2	5.920548	8.69863	2.83879781	1	1	1
349004602	2093	2	5.920548	8.849315	2.98907104	0	0	1

Table A.2. PRISM study data: Corresponding repeated doubly-interval-censored gap times between recurrent malaria parasite infections: Interval-censored data for clearance infection time U and interval-censored data for infection time T^*, subject id = 2093.

hhid	id	siteid	age	usedAL	$u_{l_{ijk}}$	$u_{r_{ijk}}$	$v_{l_{ijk}}$	$v_{r_{ijk}}$
349004602	2093	2	6.090411	1	0.23835616	0.29315068	0.99453552	1.12295082
349004602	2093	2	6.978082	1	1.12295082	1.2431694	1.72876712	1.82465753
349004602	2093	2	7.676713	1	1.82465753	1.99726027	2.51092896	2.57923497
349004602	2093	2	8.438356	1	2.57923497	2.7568306	2.7568306	2.83879781

B. Additional results for model comparison

Here, we present the model comparison results for the mixed effect accelerate failure time models (3) and (4) described in the main text of Section 3. A summary of the specification of the error distribution and the implied mixed-effect accelerated failure time model (MEAFT) of the clearance time U and the infection time T^* is shown in Table B.1. The models are parameterized according to the JAGS software (Plummer, 2003), as follows: $\kappa_U = \sigma_U^{-1}$, $\mu_U = b_{11} + b_{21} + \beta_1^{'}\mathbf{x}_1$, $\lambda_U = \exp(\mu_U)$, where μ_U and σ_U, are, respectively, location and scale parameters for $\log(U)$. Similarly, $\kappa_{T^*} = \sigma_{T^*}^{-1}$, $\mu_{T^*} = b_{12} + b_{22} + \beta_0 + \beta_2^{'}\mathbf{x}_2$, $\lambda_{T^*} = \exp(\mu_{T^*})$, where μ_{T^*} and σ_{T^*}, are, respectively, location and scale parameters for $\log(T^*)$. In the case of MEAFT models with flexible error distribution modeled as a normal mixture, the density functions of U and T^* are defined in combination with models (3) and (4), and the distribution of type (5) (Komárek et al., 2005; Komárek and Lesaffre, 2007, 2008; Bogaerts et al., 2017), as described in the main text of Section 3. Thus, $\mu_U^* = \mu_{\zeta_q} + b_{11} + b_{21} + \beta_1^{'}\mathbf{x}_1$ and $\mu_{T^*}^* = \mu_\varepsilon + b_{12} + b_{22} + \beta_2^{'}\mathbf{x}_2$.

Table B.2 displays the WAIC values for the MEAFT model. The error distributions of U and T^* are specified according to Table B.1. Q_1 and Q_2 represent the number of mixture components of U and T, as defined in model (5), Section 3 of the main text. The expected log posterior density ($elpd$) is also reported. The WAIC values of the best selected models are in bold. We noted that MEAFT models with a flexible error distribution, modeled as a Gaussian mixture (estimated error density for clearance time U and gap time T^* are shown in Figure B.1), are preferred. These models outperform all others. As a sensitivity analysis, Table B.3 shows posterior summary statistics for the parameters of the best MEAFT models across the three Scenarios, with season included in the models as a covariate. In order to check the robustness of our findings, we explicitly accounted for the number of previous infection events in the AFT model describing the conditional distribution of the clearance times, both in the absence or presence of a flexible error mixture distribution (results shown in Tables B.4 to B.5).

Table B.1. Summary of the specification of the error term distributions and implied MEAFT models of U in (3) and T^* in (4), as described in the main text of Section 3. The models are parameterized in line with JAGS software. The distributions of U and T^*, are respectively, truncated on $[u_{l_{ijk}}, u_{r_{ijk}}]$ and $[v_{l_{ijk}} - u_{ijk}, v_{r_{ijk}} - u_{ijk}]$.

Distributional form of ζ and ε	Sample $U \sim f_U(u_k\|\mathbf{\Phi})$	Sample $T^* \sim f_{T^*}(t_k^*\|\mathbf{\Theta})$
$\zeta \sim \mathrm{EV}(\mu_U, \kappa_U)$, $\varepsilon \sim \mathrm{EV}(\mu_{T^*}, \kappa_{T^*})$	$U \sim \mathrm{dweib}(\kappa_U, \lambda_U)$	$T^* \sim \mathrm{dweib}(\kappa_{T^*}, \lambda_{T^*})$
$\zeta \sim \mathrm{EV}(\mu_U, \kappa_U)$, $\varepsilon \sim \mathrm{N}(0,1)$	$U \sim \mathrm{dweib}(\kappa_U, \lambda_U)$	$T^* \sim \mathrm{dlnorm}(\mu_{T^*}, \kappa_{T^*})$
$\zeta \sim \mathrm{EV}(\mu_U, \kappa_U)$, $\varepsilon \sim \mathrm{dlogis}(\mu_{T^*}, \kappa_{T^*})$	$U \sim \mathrm{dweib}(\kappa_U, \lambda_U)$	$\log(T^*) \sim \mathrm{dlogis}(\mu_{T^*}, \kappa_{T^*})$
$\zeta \sim \mathrm{N}(0,1)$, $\varepsilon \sim \mathrm{N}(0,1)$	$U \sim \mathrm{dlnorm}(\mu_U, \kappa_{T^*})$	$T^* \sim \mathrm{dlnorm}(\mu_{T^*}, \kappa_{T^*})$
$\zeta \sim \mathrm{N}(0,1)$, $\varepsilon \sim \mathrm{EV}(\mu_{T^*}, \kappa_{T^*})$	$U \sim \mathrm{dlnorm}(\mu_U, \kappa_{T^*})$	$T^* \sim \mathrm{dweib}(\kappa_{T^*}, \lambda_{T^*})$
$\zeta \sim \mathrm{EV}(\mu_U, \kappa_U)$, $\varepsilon \sim \sum_{q=1}^{2} \omega_{T_q^*} N(\mu_{\varepsilon_q}, \sigma_{\varepsilon_q}^2)$	$U \sim \mathrm{dweib}(\kappa_U, \lambda_U)$	$\log(T^*) \sim N(\mu_{T^*}^*, \sigma_\varepsilon^2)$
$\zeta \sim \mathrm{dlogis}(\mu_U, \kappa_U)$, $\varepsilon \sim \sum_{q=1}^{2} \omega_{T_q^*} N(\mu_{\varepsilon_q}, \sigma_\varepsilon^2)$	$\log(U) \sim \mathrm{dlogis}(\mu_U, \kappa_U)$	$\log(T^*) \sim N(\mu_{T^*}^*, \sigma_\varepsilon^2)$
$\zeta \sim \mathrm{N}(0,1)$, $\varepsilon \sim \sum_{q=1}^{Q_2} \omega_{T_q^*} N(\mu_{\varepsilon_q}, \sigma_\varepsilon^2)$	$\log(U) \sim N(\mu_U^*, \sigma_\zeta^2)$	$T^* \sim \mathrm{dlnorm}(\mu_{T^*}, \kappa_{T^*})$
$\zeta \sim \sum_{q=1}^{Q_1} \omega_{U_q} N(\mu_{\zeta_q}, \sigma_\zeta^2)$, $\varepsilon \sim \mathrm{N}(0,1)$	$U \sim \mathrm{dlnorm}(\mu_U, \kappa_U)$	$\log(T^*) \sim N(\mu_{T^*}^*, \sigma_\varepsilon^2)$
$\zeta \sim \sum_{q=1}^{Q_1} \omega_{U_q} N(\mu_{\zeta_q}, \sigma_\zeta^2)$, $\varepsilon \sim \sum_{q=1}^{Q_2} \omega_{T_q^*} N(\mu_{\varepsilon_q}, \sigma_\varepsilon^2)$	$\log(U) \sim N(\mu_U^*, \sigma_{\zeta_q}^2)$	$\log(T^*) \sim N(\mu_{T^*}^*, \sigma_\varepsilon^2)$

EV(.,.), stands for Extreme-value distribution (or Gumbel density); N(.,.):= normal distribution;
dweib(.,.) is the Weibull distribution, dlnorm(.,.):= log-normal model, and dlogis denote the logistic distribution.

Table B.2. PRISM study data: Model comparison results - WAIC values for MEAFT model with specification of the error distribution of U and T^* according to Table B.1.

	Walukuba		Kihihi		Nagongera	
	elpd	WAIC	elpd	WAIC	elpd	WAIC
Error distribution of U and T^*	$\mathbf{b}_1 \sim N_2(\mathbf{0}, \Sigma_{\mathbf{b}_1})$ and $\mathbf{b}_2 \sim N_2(\mathbf{0}, \Sigma_{\mathbf{b}_2})$					
$\zeta \sim \mathrm{EV}(\mu_U, \kappa_U),\ \varepsilon \sim \mathrm{EV}(\mu_{T^*}), \kappa_{T^*})$	-1727.4	3454.8	-4713.3	9426.7	-7053.8	14107.6
$\zeta \sim \mathrm{dlogis}(\mu_U, \kappa_U),\ \varepsilon \sim \mathrm{dlogis}(\mu_{T^*}, \kappa_{T^*})$	-1766.3	3532.7	-4688.6	9377.2	-6809.4	13618.8
$\zeta \sim \mathrm{N}(0,1),\ \varepsilon \sim \mathrm{N}(0,1)$	-1763.3	3526.6	-4674.6	9349.2	-6824.2	13648.4
Eror distribution of U and T^*	$\mathbf{b}_1 \sim N_2(\mathbf{0}, \Sigma_{\mathbf{b}_1})$ and $b_2 \sim N(0, \sigma_2^2)$					
$\zeta \sim \mathrm{EV}(\mu_U, \kappa_U),\ \varepsilon \sim \mathrm{EV}(\mu_{T^*}), \kappa_{T^*})$	-1726.9	3453.8	-4733.1	9466.2	-7089.0	14178.0
$\zeta \sim \mathrm{dlogis}(\mu_U, \kappa_U),\ \varepsilon \sim \mathrm{dlogis}(\mu_{T^*}, \kappa_{T^*})$	-1775.2	3550.4	-4742.9	9485.9	-6846.6	13693.3
$\zeta \sim \mathrm{N}(0,1),\ \varepsilon \sim \mathrm{N}(0,1)$	-1776.0	3552.0	-4691.7	9383.4	-6839.9	13679.8
Mixture of normal distributions	$\mathbf{b}_1 \sim N_2(\mathbf{0}, \Sigma_{\mathbf{b}_1})$ and $b_2 \sim N(0, \sigma_2^2)$					
$Q_1 = 2, Q_2 = 2$	-1645.8	3291.6	-4381.0	8762.1	-6541.1	13082.3
$Q_1 = 2, Q_2 = 3$	-1651.4	3302.8	-4228.4	8456.8	-6461.9	12923.9
$Q_1 = 2, Q_2 = 4$	-1484.4	2968.9	-4182.5	8364.9	-6458.1	12916.2
$Q_1 = 2, Q_2 = 5$	-1491.8	2983.7	-4184.8	8369.6	-6184.3	12368.6
$Q_1 = 3, Q_2 = 2$	-1771.6	3543.2	-4391.2	8782.5	-6709.4	13418.7
$Q_1 = 3, Q_2 = 3$	-1541.0	3082.0	-4206.6	8413.2	-6494.9	12989.8
$Q_1 = 3, Q_2 = 4$	-1456.9	2913.9	-4090.3	8180.7	-6489.2	12978.3
$Q_1 = 3, Q_2 = 5$	-1452.9	2905.9	-4141.7	8283.4	-6421.5	12842.9
$Q_1 = 4, Q_2 = 2$	-1583.2	3166.3	-4359.6	8719.3	-6526.6	13053.2
$Q_1 = 4, Q_2 = 3$	-1682.5	3365.1	-4348.0	8696.1	-6495.1	12990.1
$Q_1 = 4, Q_2 = 4$	-1479.8	2959.6	-4197.6	8395.1	-6382.7	12765.4
$Q_1 = 4, Q_2 = 5$	-1405.4	2810.8	-4109.1	8281.1	-6255.3	12510.6
$Q_1 = 5, Q_2 = 2$	-1621.9	3243.8	-4385.3	8770.6	-6545.2	13090.3
$Q_1 = 5, Q_2 = 3$	-1491.1	2982.3	-4359.6	8719.3	-6347.8	12695.5
$Q_1 = 5, Q_2 = 4$	-1446.8	2893.6	-4224.5	8449.0	-6222.3	12444.6
$Q_1 = 5, Q_2 = 5$	-1368.0	**2735.9**	-4241.4	8482.8	-6242.9	12485.7
Mixture of normal distributions	$b_1 \sim N(0, \sigma_1^2)$ and $b_2 \sim N(0, \sigma_2^2)$					
$Q_1 = 2, Q_2 = 2$	-1665.9	3331.7	-4232.4	8464.8	-6584.5	13168.9
$Q_1 = 2, Q_2 = 3$	-1610.1	3220.2	-4157.1	8314.1	-6540.2	13080.4
$Q_1 = 2, Q_2 = 4$	-1482.6	2965.2	-4133.9	8267.8	-6225.0	12450.0
$Q_1 = 2, Q_2 = 5$	-1404.8	2809.7	-4095.3	8190.5	-6191.9	12383.7
$Q_1 = 3, Q_2 = 2$	-1656.5	3312.9	-4233.5	8467.0	-6571.8	13143.6
$Q_1 = 3, Q_2 = 3$	-1557.1	3114.1	-4228.2	8456.3	-6246.8	12493.7
$Q_1 = 3, Q_2 = 4$	-1492.4	2984.7	-4099.1	8198.1	-6240.7	12481.4
$Q_1 = 3, Q_2 = 5$	-1524.4	3048.9	-4082.4	8164.8	-6490.4	12980.8
$Q_1 = 4, Q_2 = 2$	-1673.4	3346.7	-4230.8	8461.7	-6572.6	13145.3
$Q_1 = 4, Q_2 = 3$	-1617.8	3235.5	-4170.5	8340.9	-6424.4	12848.7
$Q_1 = 4, Q_2 = 4$	-1582.4	3164.7	-4104.9	8209.8	-6156.5	12313.0
$Q_1 = 4, Q_2 = 5$	-1527.8	3055.4	-4066.8	**8133.6**	-6214.4	12428.8
$Q_1 = 5, Q_2 = 2$	-1671.3	3342.6	-4229.5	8459.0	-6605.3	13210.6
$Q_1 = 5, Q_2 = 3$	-1578.3	3156.6	-4154.9	8309.9	-6557.3	13114.6
$Q_1 = 5, Q_2 = 4$	-1594.6	3189.2	-4106.9	8213.8	-6566.8	13133.6
$Q_1 = 5, Q_2 = 5$	-1532.3	3064.5	-4128.7	8257.3	-6154.8	**12309.6**

Table B.3. PRISM study data: Posterior summary statistics for the parameters of the best selected MEAFT models across the three Scenarios, with error distribution modeled as a mixture of normal distributions. Posterior mean, posterior standard deviation in parentheses, and 95% HPDI. As a sensitivity analysis season included as a covaria in the model.

	Scenario 1		Scenario 2		Scenario 3	
Parameter	Estimate (SD)	95%CI	Estimate (SD)	95%CI	Estimate (SD)	95%CI
Walukuba: $\mathbf{b}_1 \sim N_2(\mathbf{0}, \Sigma_{\mathbf{b}_1})$, $\mathbf{b}_2 \sim N(0, \sigma_2^2)$, $\rho_2 = 1.000$						
β_{20}	-1.989 (0.117)	[-2.232, -1.768]	-1.307 (0.115)	[-1.527, -1.088]	-4.113 (0.319)	[-4.681, -3.415]
β_{age}	0.003 (0.003)	[-0.004, 0.009]	0.010 (0.004)	[0.001, 0.017]	0.029 (0.013)	[0.004, 0.051]
β_{gender} (male)	-0.037 (0.086)	[-0.208, 0.134]	-0.004 (0.102)	[-0.213, 0.182]	0.279 (0.351)	[-0.411. 0.845]
β_{AL}	1.419 (0.075)	[1.275, 1.563]	1.344 (0.089)	[1.173, 1.517]	1.419 (0.117)	[1.198, 1.648]
β_{season} (rainy)	-0.047 (0.078)	[-0.199, 0.103]	-0.488 (0.088)	[-0.654, -0.317]	-0.455 (0.106)	[-0.668, -0.249]
σ_1^2	0.126 (0.044)	[0.054,0.225]	0.193 (0.066)	[0.088, 0.344]	3.085 (0.833)	[1.728, 4.864]
σ_2^2	0.071 (0.038)	[0.014, 0.158]	0.021 (0.015)	[0.004, 0.060]	6.333 (0.372)	[5.852, 7.142]
ρ_1	0.277 (0.229)	[0.009, 0.892]	0.493 (0.131)	[0.236, 0.749]	0.946 (0.054)	[0.804, 0.994]
Logistic model						
γ_0					1.055 (0.342)	[0.439, 1.787]
γ_{geEntry}					0.009 (0.015)	[-0.021, 0.038]
γ_{gender} (male)					0.587 (0.109)	[0.373, 0.801]
Kihihi: $\mathbf{b}_1 \sim N(0, \sigma_1^2)$, $\mathbf{b}_2 \sim N(0, \sigma_2^2)$, $\rho_1 = \rho_2 = 1.000$						
β_{20}	-2.682 (0.086)	[-2.859, -2.601]	-2.623 (0.088)	[-2.786, -2.430]	-2.490 (0.102)	[-2.694, -2.294]
β_{age}	0.015 (0.002)	[0.011, 0.019]	0.023 (0.003)	[0.018, 0.028]	0.022 (0.003)	[0.018, 0.027]
β_{gender} (male)	-0.102 (0.062)	[-0.221, 0.011]	-0.208 (0.076)	[-0.357, -0.0620]	-0.188 (0.077)	[-0.365, -0.059]
β_{AL}	1.466 (0.060)	[1.355, 1.587]	1.570 (0.063)	[1.444, 1.698]	1.446 (0.064)	[1.320, 1.567]
β_{season} (rainy)	-0.236 (0.054)	[-0.353, -0.137]	-0.165 (0.058)	[-0.281, -0.054]	-0.178 (0.065)	[-0.305. -0.051]
σ_1^2	0.246 (0.057)	[0.150, 0.368]	0.449 (0.080)	[0.310, 0.625]	0.824 (0.159)	[0.554, 1.114]
σ_2^2	0.040 (0.005)	[0.003, 0.102]	0.053 (0.030)	[0.012, 0.118]	0.020 (0.017)	[0.002, 0.068]
Logistic model						
γ_0					4.363 (0.199)	[3.988, 4.762]
γ_{geEntry}					-0.021 (0.007)	[-0.033, -0.007]
γ_{gender} (male)					1.078 (0.331)	[0.448, 1.751]
Nagongera: $\mathbf{b}_1 \sim N(0, \sigma_1^2)$, $\mathbf{b}_2 \sim N(0, \sigma_2^2)$, $\rho_1 = \rho_2 = 1.000$						
β_{20}	-3.160 (0.066)	[-3.286, -3.030]	-3.297 (0.061)	[-3.410, -3.187]	-3.311 (0.073)	[-3.450, -3.156]
β_{age}	0.024 (0.002)	[0.019, 0.028]	0.030 (0.002)	[0.025, 0.034]	0.031 (0.002)	[0.026, 0.035]
β_{gender} (male)	-0.281 (0.059)	[-0.405, -0.174]	-0.318 (0.065)	[-0.440, -0.200]	-0.303 (0.065)	[-0.430, -0.172]
β_{AL}	1.536 (0.062)	[1.417, 1.662]	1.953 (0.050)	[1.863, 2.053]	1.925 (0.057)	[1.810, 2.037]
β_{season} (rainy)	-0.174 (0.045)	[-0.256, -0.087]	-0.081 (0.044)	[-0.167, 0.003]	-0.066 (0.048)	[-0.162. 0.029]
σ_1^2	0.016 (0.013)	[0.002, 0.047]	0.014 (0.008)	[0.002, 0.032]	0.038 (0.014)	[0.016, 0.066]
σ_2^2	0.075 (0.027)	[0.023, 0.128]	0.121 (0.032)	[0.069, 0.185]	0.115 (0.029)	[0.061, 0.182]
Logistic model						
γ_0					4.130 (0.275)	[3.637, 4.706]
γ_{geEntry}					0.106 (0.041)	[0.042, 0.205]
γ_{gender} (male)					0.713 (0.329)	[0.085, 1.363]

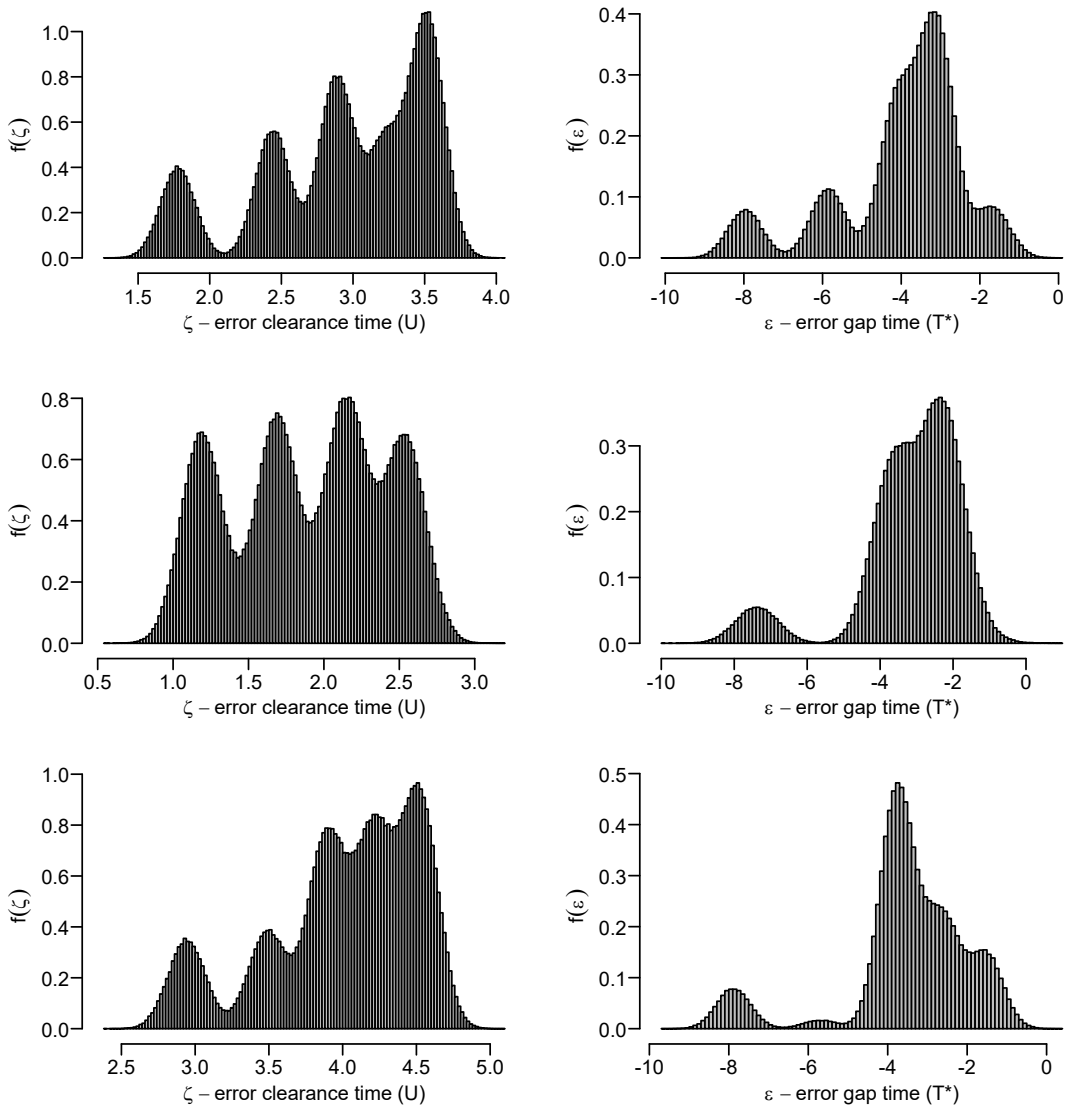

Figure B.1. PRISM study data: Estimated error density for clearance time U (left column) and gap time T^* (right column) obtained by fitting model Scenario 3. Top panel: Walukuba, Middle panel: Kihihi, Bottom: Nagongera.

Table B.4. PRISM study data: Posterior mean, posterior standard deviation in parentheses, and 95% credible interval obtained by fitting the MEAFT models with error distribution as homogeneous normal distribution. Number of previous event included in the model of clearance time U.

	Scenario 1		Scenario 2		Scenario 3	
Parameter	Estimate (SD)	95%CI	Estimate (SD)	95%CI	Estimate (SD)	95%CI
Walukuba: $\mathbf{b}_1 \sim N_2(\mathbf{0}, \Sigma_{\mathbf{b}_1})$, $\mathbf{b}_2 \sim N(0, \sigma_2^2)$, $\rho_2 = 1.000$						
β_{20}	-1.778 (0.149)	[-2.062, -1.477]	-1.295 (0.170)	[-1.609, -0.956]	-3.896 (0.410)	[-4.758, -3.062]
β_{age}	0.007 (0.005)	[-0.003, 0.017]	0.014 (0.005)	[0.004, 0.025]	0.024 (0.014)	[-0.003, 0.055]
β_{gender}	-0.086 (0.133)	[-0.350, 0.171]	-0.109 (0.142)	[-0.397, 0.163]	0.405 (0.419)	[-0.374, 1.258]
β_{AL}	1.364 (0.115)	[1.141, 1.587]	1.365 (0.124)	[1.114, 1.607]	1.505 (0.149)	[1.215, 1.804]
β^*_{Kevents}	2.091 (0.929)	[0.584, 4.178]	3.794 (1.685)	[1.092, 7.583]	5.189 (1.937)	[1.596, 9.562]
σ_1^2	0.333 (0.115)	[0.153, 0.588]	0.512 (0.145)	[0.301, 0.896]	3.00 (0.847)	[1.581, 4.820]
σ_2^2	0.075 (0.070)	[1e-04, 0.235]	0.071 (0.058)	[0.007, 0.222]	9.819 (1.404)	[7.314, 12.875]
ρ_1	0.449 (0.082)	[0.288, 0.610]	0.491 (0.122)	[0.252, 0.730]	0.735 (0.121)	[0.486, 0.960]
Logistic regression						
γ_0					1.621 (0.366)	[0.883, 2.303]
γ_{geEntry}					0.003 (0.015)	[-0.027, 0.035]
γ_{gender}					0.847 (0.443)	[-0.002, 1.736]
Kihihi: $\mathbf{b}_1 \sim N(0, \sigma_1^2)$, $\mathbf{b}_2 \sim N(0, \sigma_2^2)$, $\rho_1 = \rho_2 = 1.000$						
β_{20}	-2.892 (0.104)	[-3.094, -2.687]	-2.854 (0.175)	[-3.207, -2.512]	-2.854 (0.175)	[-3.207, -2.512]
β_{age}	0.18 (0.003)	[0.012, 0.023]	0.023 (0.003)	[0.018, 0.029]	0.020 (0.005)	[0.008, 0.028]
β_{gender}	-0.149 (0.085)	[-0.322, 0.015]	-0.240 (0.087)	[-0.420, -0.075]	-0.237 (0.118)	[-0.470, 0.001]
β_{AL}	1.690 (0.069)	[1.557, 1.828]	1.720 (0.089)	[1.515, 1.874]	1.680 (0.083)	[1.523, 1.851]
β^*_{Kevents}	3.962 (0.869)	[2.646, 6.001]	3.751 (1.799)	[-0.023, 6.384]	5.388 (1.637)	[2.987, 8.948]
σ_1^2	0.396 (0.083)	[0.260, 0.576]	0.582 (0.117)	[0.385, 0.845]	1.181 (0.340)	[0.607, 1.935]
σ_2^2	0.089 (0.039)	[0.019, 0.173]	0.156 (0.046)	[0.075, 0.253]	0.399 (0.414)	[0.098, 1.842]
Logistic regression						
γ_0					4.316 (0.235)	[3.870, 4.801]
γ_{geEntry}					-0.021 (0.008)	[-0.035, -0.005]
γ_{gender}					1.022 (0.329)	[0.374, 1.680]
Nagongera: $\mathbf{b}_1 \sim N(0, \sigma_1^2)$, $\mathbf{b}_2 \sim N(0, \sigma_2^2)$, $\rho_1 = \rho_2 = 1.000$						
β_{20}	-3.301 (0.061)	[-3.423, -3.179]	-3.352 (0.071)	[-3.485, -3.209]	-3.400 (0.075)	[-3.47, -3.258]
β_{age}	0.025 (0.002)	[0.021, 0.029]	0.029 (0.002)	[0.025, 0.034]	0.031 (0.003)	[0.025, 0.035]
β_{gender}	-0.345 (0.062)	[-0.469, -0.224]	-0.354 (0.072)	[-0.502, -0.218]	-0.338 (0.079)	[-0.484, -0.182]
β_{AL}	1.706 (0.050)	[1.612, 1.806]	2.017 (0.055)	[1.910, 2.124]	2.001 (0.053)	[1.904, 2.110]
β^*_{Kevents}	3.136 (0.866)	[1.295, 4.304]	4.770 (0.859)	[3.137, 6.475]	7.598 (0.879)	[6.082, 9.652]
σ_1^2	0.035 (0.019)	[0.007, 0.078]	0.048 (0.025)	[0.012, 0.106]	0.031 (0.020)	[0.006, 0.083]
σ_2^2	0.125 (0.033)	[0.070, 0.200]	0.160 (0.040)	[0.090, 0.242]	0.202 (0.047)	[0.110, 0.293]
Logistic regression						
γ_0					4.251 (0.275)	[3.714, 4.816]
γ_{geEntry}					0.096 (0.037)	[0.037, 0.179]
γ_{gender}					0.784 (0.366)	[0.110, 1.544]

β^*_{Kevents}:= number of previous event included in the model of clearance time U.

Table B.5. PRISM study data: Posterior mean, posterior standard deviation in parentheses, and 95% credible interval obtained by fitting the MEAFT models with error distribution as normal mixture. Number of previous event included in the model of clearance time U.

	Scenario 1		Scenario 2		Scenario 3	
Parameter	Estimate (SD)	95%CI	Estimate (SD)	95%CI	Estimate (SD)	95%CI
Walukuba: $\mathbf{b}_1 \sim N_2(\mathbf{0}, \Sigma_{\mathbf{b}_1})$, $\mathbf{b}_2 \sim N(0, \sigma_2^2)$, $\rho_2 = 1.000$						
β_{20}	-2.104 (0.159)	[-2.378, -1.758]	-1.457 (0.162)	[-1.745, -1.103]	-3.747 (0.363)	[-4.318, -2.860]
β_{age}	0.002 (0.004)	[-0.006, 0.009]	0.010 (0.004)	[0.001, 0.017]	0.018 (0.011)	[-0.007, 0.039]
β_{gender}	-0.058 (0.115)	[-0.313, 0.145]	-0.056 (0.109)	[-0.270, 0.149]	0.297 (0.329)	[-0.328, 0.922]
β_{AL}	1.402 (0.089)	[1.221, 1.566]	1.342 (0.098)	[1.144, 1.532]	1.418 (0.126)	[1.185, 1.670]
β^*_{Kevents}	5.212 (2.423)	[0.610, 9.910]	4.775 (2.649)	[-0.175, 10.156]	6.127 (1.516)	[3.156, 9.098]
σ_1^2	0.121 (0.053)	[0.033, 0.238]	0.302 (0.082)	[0.158, 0.482]	2.404 (0.862)	[0.702, 3.939]
σ_2^2	0.123 (0.051)	[0.033, 0.232]	0.056 (0.048)	[0.005, 0.172]	8.323 (1.309)	[6.002, 11.110]
ρ_1	0.440 (0.078)	[0.287, 0.593]	0.448 (0.111)	[0.230, 0.666]	0.400 (0.102)	[0.200, 0.600]
Logistic regression						
γ_0					1.635 (0.336)	[1.089, 2.379]
γ_{geEntry}					1e-05 (0.013)	[-0.024, 0.025]
γ_{gender}					0.668 (0.378)	[-0.081, 1.398]
Kihihi: $\mathbf{b}_1 \sim N(0, \sigma_1^2)$, $\mathbf{b}_2 \sim N(0, \sigma_2^2)$, $\rho_1 = \rho_2 = 1.000$						
β_{20}	-2.818 (0.091)	[-2.988, -2.627]	-2.700 (0.106)	[-2.912, -2.486]	-2.791 (0.156)	[-3.101, -2.478]
β_{age}	0.013 (0.002)	[0.009, 0.018]	0.023 (0.003)	[0.017, 0.028]	0.019 (0.004)	[0.011, 0.0267]
β_{gender}	-0.125 (0.066)	[-0.256, 0.009]	-0.223 (0.079)	[-0.378, -0.065]	-0.249 (0.092)	[-0.428, -0.074]
β_{AL}	1.429 (0.064)	[1.296, 1.546]	1.562 (0.072)	[1.414, 1.698]	1.424 (0.082)	[1.270, 1.588]
β^*_{Kevents}	8.642 (2.118)	[4.775, 13.068]	9.006 (1.835)	[5.576, 12.586]	3.812 (1.628)	[0.792, 7.013]
σ_1^2	0.269 (0.059)	[0.172, 0.402]	0.520 (0.101)	[0.347, 0.739]	1.001 (0.291)	[0.509, 1.640]
σ_2^2	0.046 (0.023)	[0.010, 0.099]	0.124 (0.039)	[0.054, 0.209]	0.159 (0.104)	[0.044, 0.467]
Logistic regression						
γ_0					4.373 (0.229)	[3.958, 4.850]
γ_{ageEntry}					-0.023 (0.008)	[-0.037, -0.008]
γ_{gender}					0.964 (0.324)	[0.359, 1.619]
Nagongera: $\mathbf{b}_1 \sim N(0, \sigma_1^2)$, $\mathbf{b}_2 \sim N(0, \sigma_2^2)$, $\rho_1 = \rho_2 = 1.000$						
β_{20}	-3.415 (0.079)	[-3.559, -3.259]	-3.373 (0.070)	[-3.518, -3.245]	-3.378 (0.077)	[-3.522, -3.219]
β_{age}	0.024 (0.002)	[0.019, 0.029]	0.029 (0.002)	[0.024, 0.034]	0.030 (0.003)	[0.025, 0.035]
β_{gender}	-0.272 (0.056)	[-0.376, -0.160]	-0.355 (0.073)	[-0.496, -0.210]	-0.339 (0.074)	[-0.489, -0.191]
β_{AL}	1.469 (0.050)	[1.373, 1.562]	2.043 (0.058)	[1.932, 2.159]	1.983 (0.063)	[1.852, 2.101]
β^*_{Kevents}	6.323 (2.218)	[2.080, 9.768]	4.814 (2.065)	[1.305, 8.251]	2.114 (0.909)	[0.770, 3.951]
σ_1^2	0.032 (0.018)	[0.008, 0.074]	0.048 (0.029)	[0.011, 0.119]	0.035 (0.022)	[0.007, 0.087]
σ_2^2	0.093 (0.025)	[0.052, 0.150]	0.156 (0.036)	[0.088, 0.229]	0.199 (0.051)	[0.109, 0.311]
Logistic regression						
γ_0					4.239 (0.283)	[3.688, 4.817]
γ_{geEntry}					0.097 (0.038)	[0.036, 0.183]
γ_{gender}					0.764 (0.355)	[0.088, 1.501]

β^*_{Kevents}:= number of previous event included in the model of clearance time U.

C. Model convergence diagnostic - Walukuba

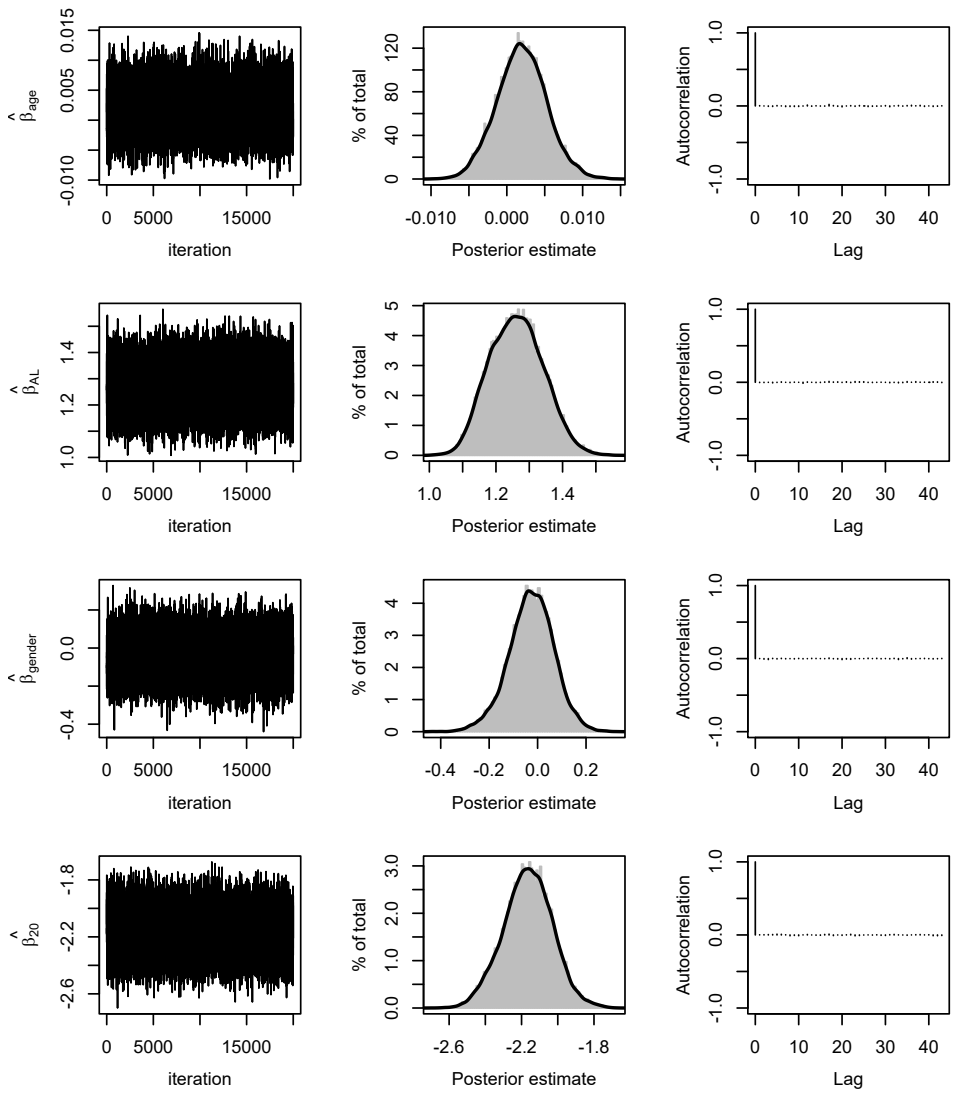

Figure C.1. Model convergence diagnostics: Trace plots (left panel), posterior distribution (middle panel), and auto-correlation plots (right panel). The MCMC algorithm was run for 120,000 iterations to ensure convergence and excluding the initial 20,000 iterations, and thinned by a factor of 6 (to decrease auto-correlation of the chain) so that the remaining 20,000 samples (single chain) were used to estimate the posterior distributions of the model parameters - **Walukuba**.

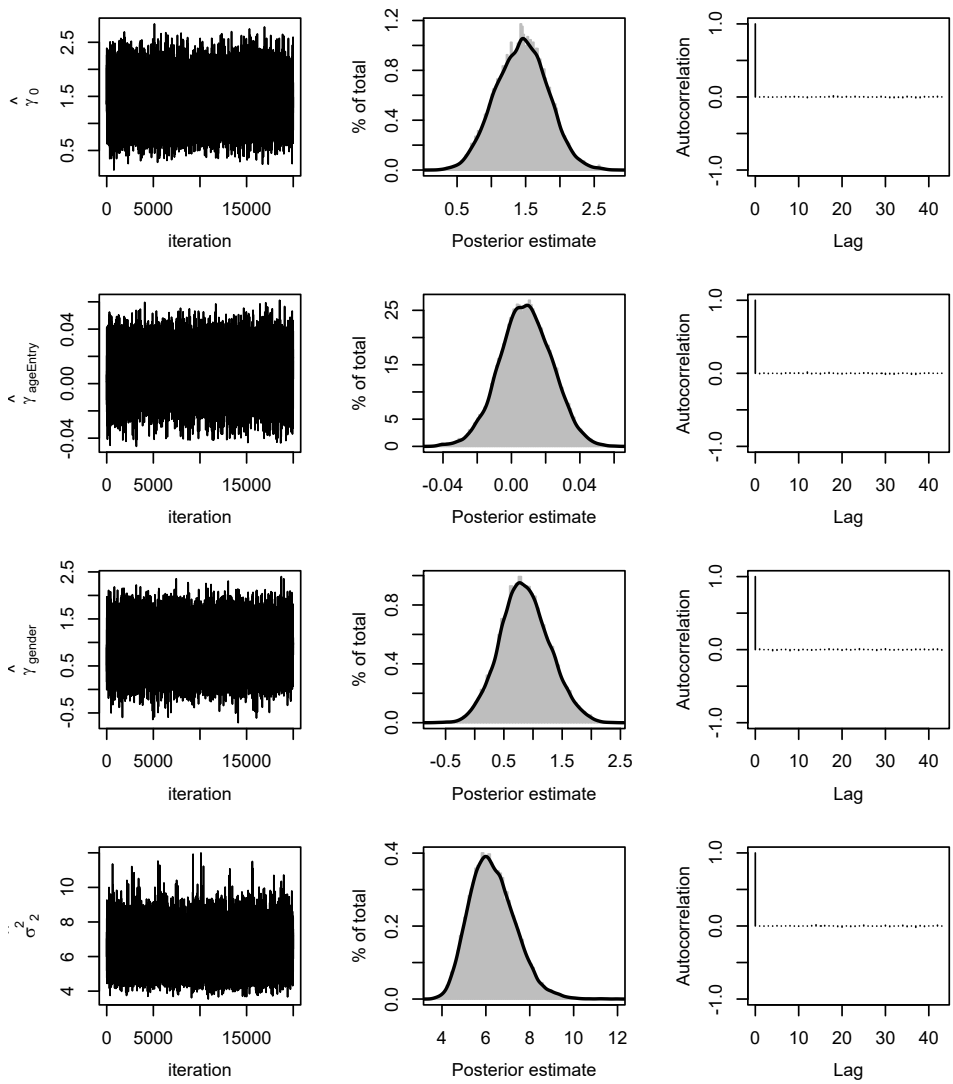

Figure C.2. Model convergence diagnostics: Trace plots (left panel), posterior distribution (middle panel), and auto-correlation plots (right panel). The MCMC algorithm was run for 120,000 iterations to ensure convergence and excluding the initial 20,000 iterations, and thinned by a factor of 6 (to decrease auto-correlation of the chain) so that the remaining 20,000 samples (single chain) were used to estimate the posterior distributions of the model parameters - **Walukuba**.

D. Model convergence diagnostic - Kihihi

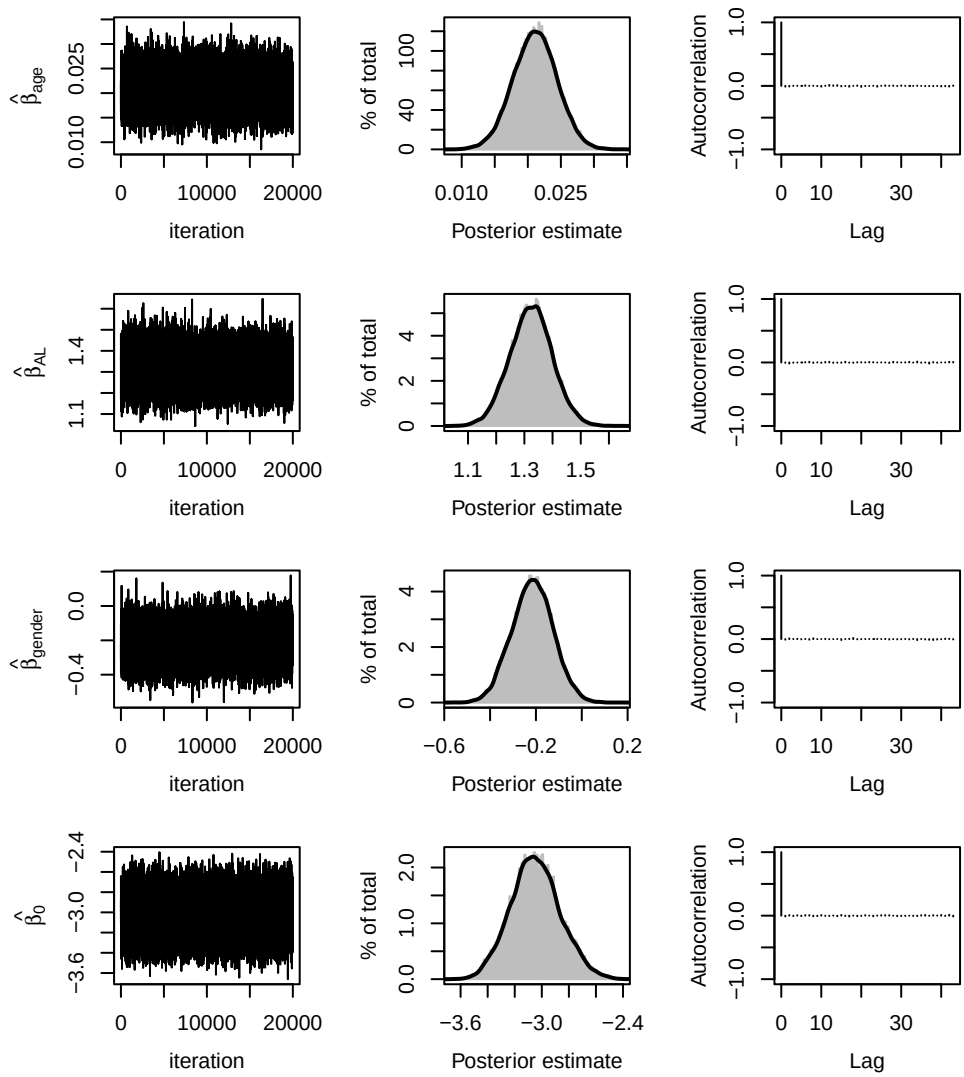

Figure D.1. Model convergence diagnostics: Trace plots (left panel), posterior distribution (middle panel), and auto-correlation plots (right panel). The MCMC algorithm was run for 120,000 iterations to ensure convergence and excluding the initial 20,000 iterations, and thinned by a factor of 6 (to decrease auto-correlation of the chain) so that the remaining 20,000 samples (single chain) were used to estimate the posterior distributions of the model parameters - **Kihihi**.

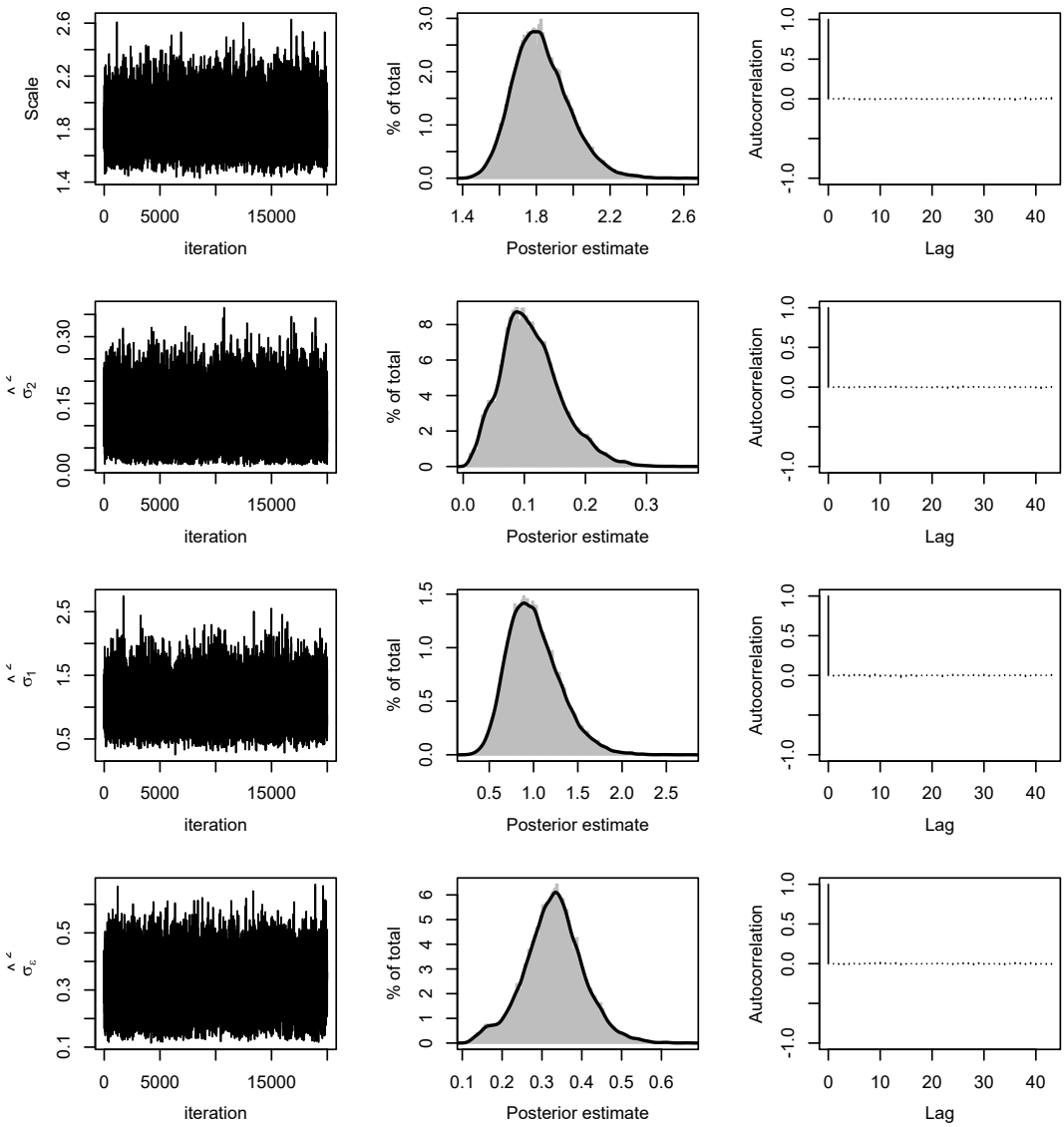

Figure D.2. Model convergence diagnostics: Trace plots (left panel), posterior distribution (middle panel), and auto-correlation plots (right panel). The MCMC algorithm was run for 120,000 iterations to ensure convergence and excluding the initial 20,000 iterations, and thinned by a factor of 6 (to decrease auto-correlation of the chain) so that the remaining 20,000 samples (single chain) were used to estimate the posterior distributions of the model parameters - **Kihihi**.

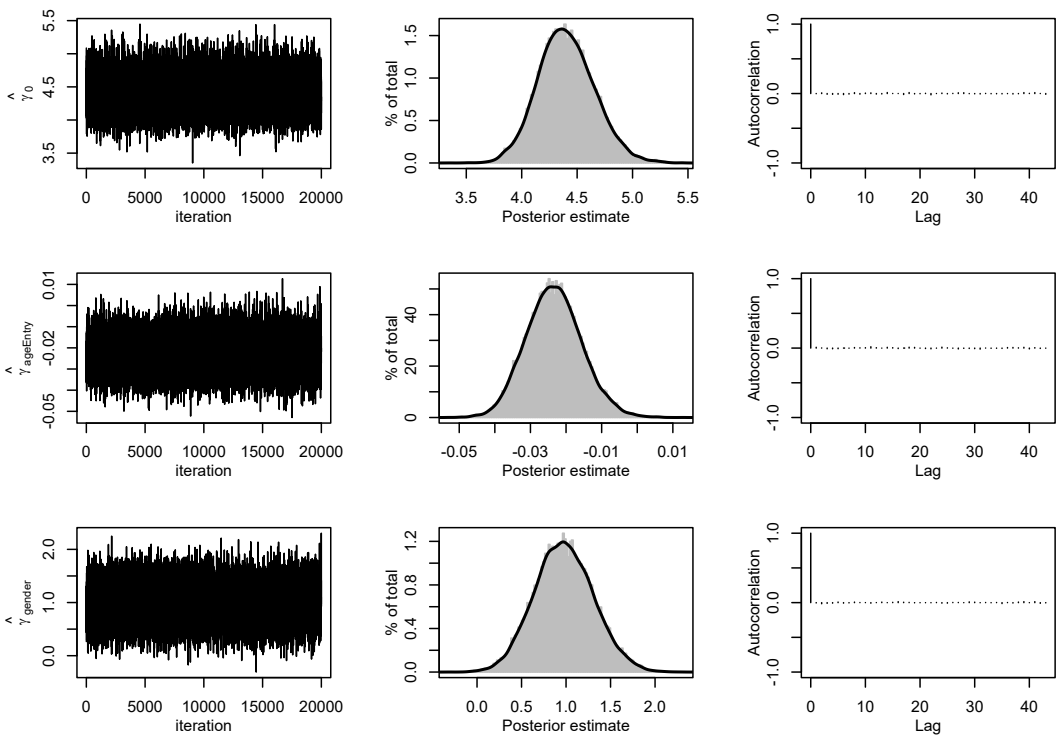

Figure D.3. Model convergence diagnostics: Trace plots (left panel), posterior distribution (middle panel), and auto-correlation plots (right panel). The MCMC algorithm was run for 120,000 iterations to ensure convergence and excluding the initial 20,000 iterations, and thinned by a factor of 6 (to decrease auto-correlation of the chain) so that the remaining 20,000 samples (single chain) were used to estimate the posterior distributions of the model parameters - **Kihihi**.

E. Model convergence diagnostic - Nagongera

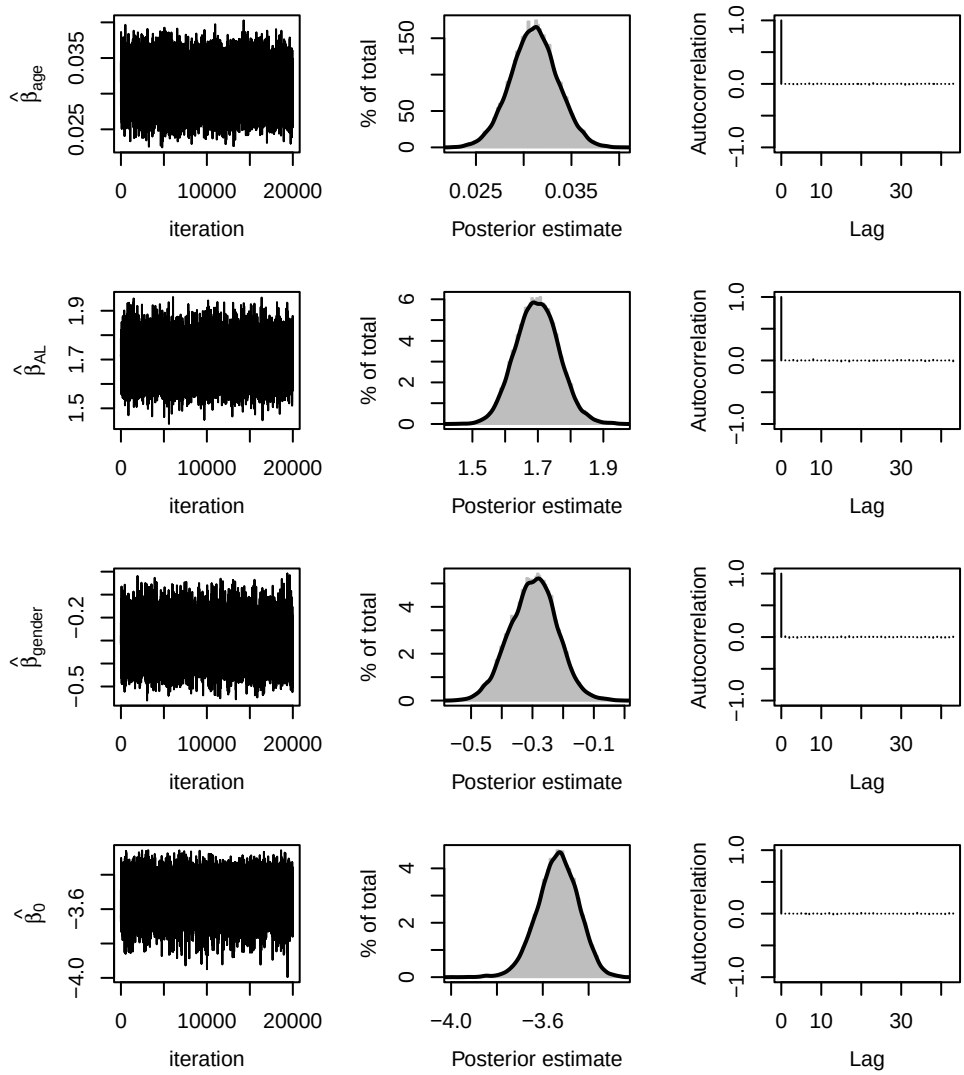

Figure E.1. Model convergence diagnostics: Trace plots (left panel), posterior distribution (middle panel), and auto-correlation plots (right panel). The MCMC algorithm was run for 120,000 iterations to ensure convergence and excluding the initial 20,000 iterations, and thinned by a factor of 6 (to decrease auto-correlation of the chain) so that the remaining 20,000 samples (single chain) were used to estimate the posterior distributions of the model parameters - **Nagongera**.

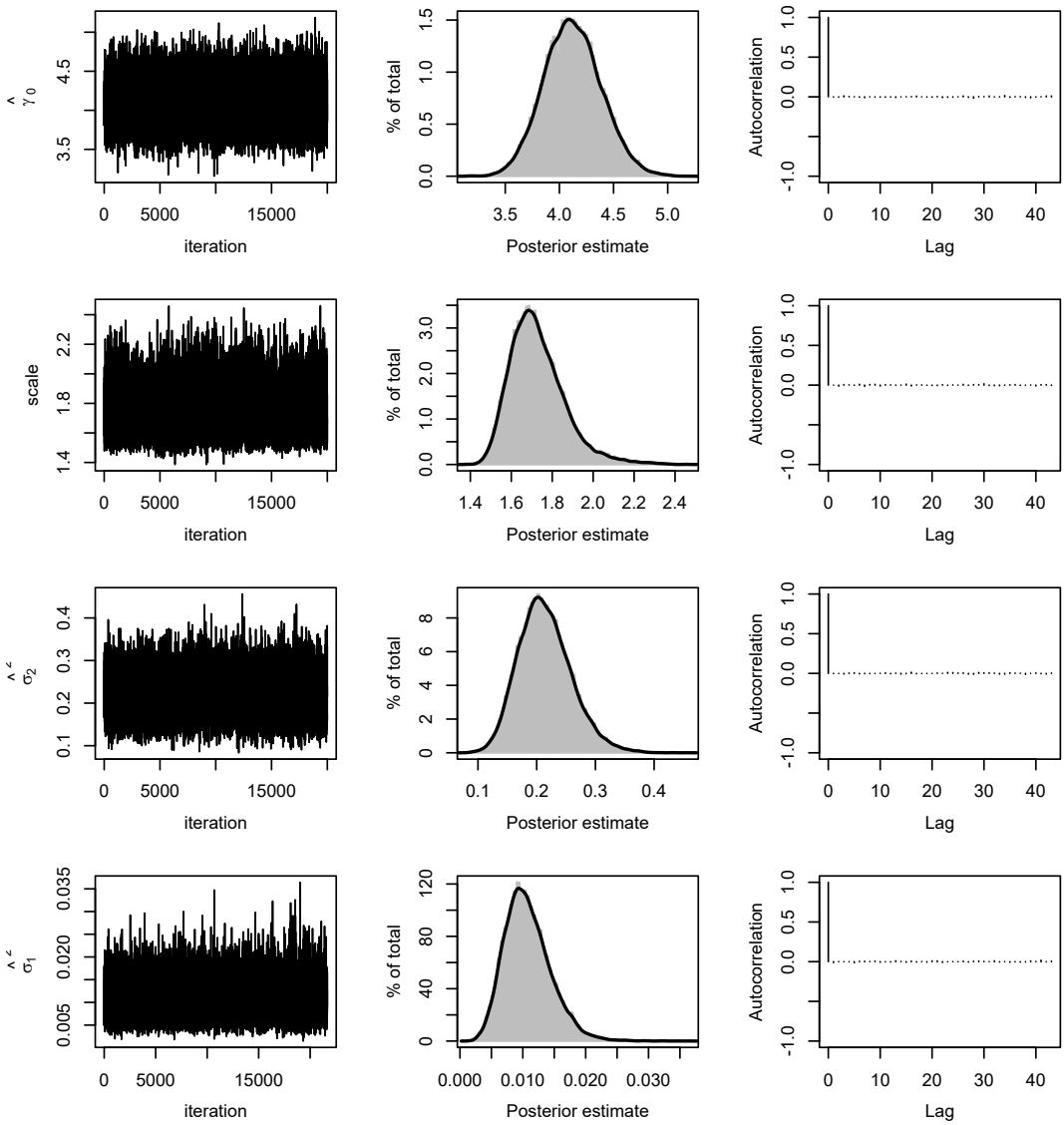

Figure E.2. Model convergence diagnostics: Trace plots (left panel), posterior distribution (middle panel), and auto-correlation plots (right panel). The MCMC algorithm was run for 120,000 iterations to ensure convergence and excluding the initial 20,000 iterations, and thinned by a factor of 6 (to decrease auto-correlation of the chain) so that the remaining 20,000 samples (single chain) were used to estimate the posterior distributions of the model parameters - **Nagongera**.

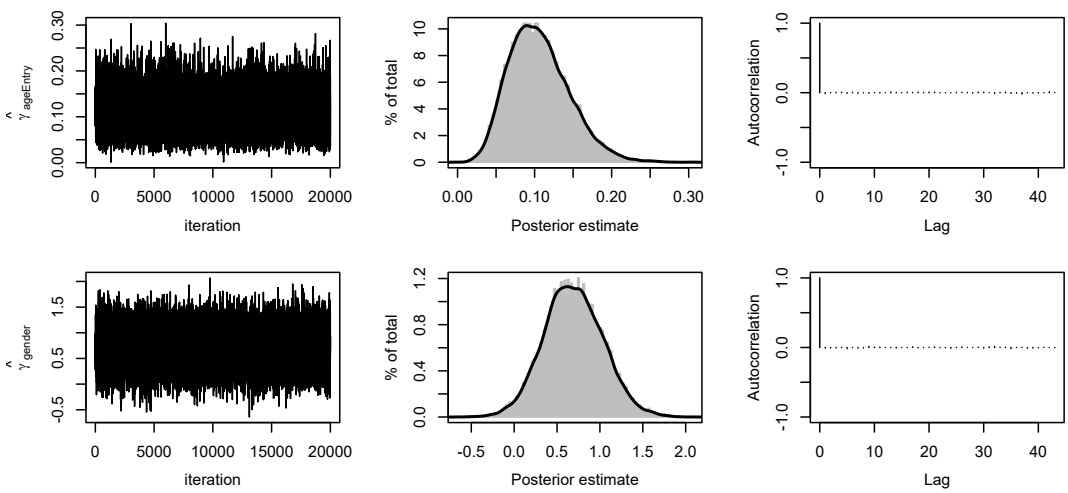

Figure E.3. Model convergence diagnostics: Trace plots (left panel), posterior distribution (middle panel), and auto-correlation plots (right panel). The MCMC algorithm was run for 120,000 iterations to ensure convergence and excluding the initial 20,000 iterations, and thinned by a factor of 6 (to decrease auto-correlation of the chain) so that the remaining 20,000 samples (single chain) were used to estimate the posterior distributions of the model parameters - **Nagongera**.

F. R implementation of the Bayesian parametric mixed effect accelerated failure time model

The Bayesian methods for the MEAFT models described in the main text of Section 3, and applied to the PRISM data, are implemented in the statistical software package R, version 4.5.0 (R Development Core Team, 2025). We provide the R program to fit the proposed models in the analysis of gap times between recurrent event data, when there are individuals present without experiencing the event of interest during the study period. To run the R code coded here, one requires the data to be organized in a linear mixed random effects model style of data input, as explained in Section A.

```
writeLines("
model{
for(i in 1:Obs){
# Random variable U
logUl[i] <- log(Ul[i])
logUr[i] <- I[i]*log(Ur[i])+(1-I[i])*log(Ur[i]+0.01)
lpU[i] <- eta1[hhid[i]]+eta2[hhid[id[i]]]
epsilonU[i] ~ dnorm(muEU[i],tau.epsilonU)
RU1[i] ~ dcat(PU[1:2])
RU[i] <- RU1[i] - 1
muEU[i] <- muEU1*RU[i] + muEU2*(1-RU[i])
Iu[i] ~ dinterval(timeu[i],c(logUl[i],logUr[i]))
timeu[i] ~ dnorm(mulogU[i],tau.epsilonU)
mulogU[i] <- muEU[i]+lpU[i]
#--------------------------------
# Gap time T = V-U
u[i] <- I[i]*exp(timeu[i])+(1-I[i])*Ur[i]
tL[i] <- log(Vl[i] - u[i])
tR[i] <- log(Vr[i] - u[i])
epsilonT[i] ~ dnorm(mu.epsilonT[i],tau.epsilonT)
R1[i] ~ dcat(PT[1:2])
R[i] <- R1[i] - 1
mu.epsilonT[i] <- muET1*R[i] + muET2*(1-R[i])
It[i] ~ dinterval(timet[i],c(tL[i],tR[i]))
timet[i] ~ dnorm(muT[i],tau.epsilonT)
random[i] <- b1[hhid[i]]+b2[hhid[id[i]]]
lpT[i] <- beta[1]*age[i]+beta[2]*usedAL[i]+beta[3]*sex[i]+random[i]
muT[i] <- mu.epsilonT[i]+lpT[i]
#-------------------------------------------------------------------
# Hazard
#-------------------------------------------------------------------
}
# Household random effect
for(k in 1:HH){
```

```
eta1[k] ~ dnorm(0,tau.eta1) # random effect HH for U
b1[k] ~ dnorm(0,tau.b1)   # random effect HH for T
}
# Child random effect
for(q in 1:Subs){
eta2[q] ~ dnorm(0,tau.eta2)# individual random effect for U
b2[q] ~ dnorm(0,tau.b2)    # individual random effect for T=V-U
}
#-------------------------------------------------------------------
# prior distributions for beta (T) and gama (U)
for(r in 1:3){beta[r] ~ dnorm(mu0[r],tau0[r])}
for(r1 in 1:3){
mu0[r1] ~ dnorm(0,1.0E-5)
tau0[r1] ~ dgamma(1.0E-2,1.0E-2)
}
#-----------------------------------------------------------------
muET1 ~ dnorm(-1.837,0.01)
muET2 ~ dnorm(-1.837,0.01)T(muET1,)
muEU1 ~ dnorm(-0.287,0.01)
muEU2 ~ dnorm(-0.287,0.01)T(muEU1,)
tau.epsilonT ~ dgamma(1.5,tau.epsT)
tau.epsilonU ~ dgamma(1.8,tau.epsU)
tau.epsT ~ dgamma(0.75,100)
tau.epsU ~ dgamma(0.81,100)
sigma2.tau.epsilonU <- 1/tau.epsilonU
sigma2.tau.epsilonT <- 1/tau.epsilonT
PT[1:2] ~ ddirch(c(1,1)) # non-informative dirichlet prior
PU[1:2] ~ ddirch(c(1,1)) # non-informative dirichlet prior
tau.eta1 ~ dgamma(1.0E-2,1.0E-2)
tau.eta2 ~ dgamma(1.0E-2,1.0E-2)
tau.b1 ~ dgamma(1.0E-2,1.0E-2)
tau.b2 ~ dgamma(1.0E-2,1.0E-2)
sigma2b1 <- 1/tau.b1
sigma2b2 <- 1/tau.b2
sigma2HHold   <- exp(sigma2b1) - 1
sigma2Subject <- exp(sigma2b2) - 1
intercept <- PT[1]*muET1+PT[2]*muET2
term1 <- PT[1]*(sigma2.tau.epsilonT+pow(muET1-intercept,2))
term2 <- PT[2]*(sigma2.tau.epsilonT+pow(muET2-intercept,2))
scale <- sqrt(term1+term2)
} ", con="FinalModel.txt")
```

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